



JavaScript Functions: Declaring, Using, With/Without Parameters, With/Without Return, Arrow Functions

1. How to Declare Functions

There are two main styles:

Function Declaration

```
function greet() {  
  console.log("Hello!");  
}  
greet(); // calls the function
```

Function Expression

```
const greet = function() {  
  console.log("Hello!");  
};  
greet();
```

Arrow Function Expression

```
const greet = () => {  
  console.log("Hello!");  
};  
greet();
```

Arrow functions can be used for shorter syntax and are often used as callbacks. [\[1\]](#) [\[2\]](#) [\[3\]](#)

2. Why Use Functions?

- **Code Reusability:** Write code once, reuse it many times.
- **Organization and Modularity:** Break programs into logical parts.
- **Parameterization:** Process different data via arguments.
- **Simplifies Maintenance:** Isolates bugs, makes updates easier.
- **Scope Management:** Variables inside a function are local, preventing conflicts. [\[4\]](#) [\[5\]](#) [\[1\]](#)

3. Functions with Parameters

Declaration

```
function add(a, b) {  
  return a + b;  
}  
console.log(add(2, 3)); // Output: 5
```

- Parameters are listed in parentheses.
- Arguments are values you pass when calling. [\[6\]](#) [\[7\]](#)

Default Parameter Value

```
function greet(name, greeting = "Hello") {  
  console.log(greeting + ", " + name + "!");  
}  
greet("Alice"); // Uses default greeting  
greet("Bob", "Goodbye"); // Uses custom greeting
```

4. Functions without Parameters

```
function showMessage() {  
  console.log("Welcome!");  
}  
showMessage();
```

- No parentheses contents means no input needed; function just runs its code. [\[8\]](#) [\[9\]](#)

5. Functions With `return`

With Return

```
function square(x) {  
  return x * x;  
}  
let result = square(4); // result is 16
```

- `return` sends output value to the caller.
- If omitted, the function returns `undefined` by default. [\[10\]](#) [\[11\]](#) [\[12\]](#) [\[13\]](#)

Without Return

```
function logHello() {  
  console.log("Hello!");  
}
```

```
}  
let msg = logHello(); // msg is undefined
```

6. Arrow Functions (Parameters, Return, No Parameters)

With parameters, with return:

```
const sum = (a, b) => a + b;  
console.log(sum(5, 7)); // returns 12
```

Without parameters:

```
const sayHello = () => 'Hello World';  
console.log(sayHello()); // returns "Hello World"
```

With block body and no explicit return:

```
const double = x => {  
  return x * 2;  
};  
console.log(double(10)); // returns 20
```

No parameters, block statement:

```
const greet = () => {  
  console.log('Welcome!');  
};  
greet();
```

Arrow functions allow concise, readable syntax, especially for simple calculations or callbacks.
[\[2\]](#) [\[14\]](#) [\[3\]](#)

Summary Table

Syntax Example	With Parameters	Without Parameters	With Return	Without Return	Arrow Function
<code>function fn(a, b) { ... }</code>	Yes	Yes/No	Yes/No	Yes/No	No
<code>const fn = function() { }</code>	Yes/No	Yes/No	Yes/No	Yes/No	No
<code>const fn = () => { ... }</code>	Yes/No	Yes/No	Yes/No	Yes/No	Yes

All styles let you mix parameters and return statements as needed. Arrow functions offer the shortest form for simple functions.

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2. <https://www.freecodecamp.org/news/javascript-arrow-functions-in-depth/>
3. https://www.w3schools.com/js/js_arrow_function.asp
4. <https://eitca.org/web-development/eitc-wd-jsf-javascript-fundamentals/functions-in-javascript/introducing-functions/examination-review-introducing-functions/what-is-the-advantage-of-using-functions-in-javascript/>
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6. https://www.tutorialspoint.com/javascript/javascript_function_parameters.htm
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10. https://www.w3schools.com/jsref/jsref_return.asp
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12. <https://www.geeksforgeeks.org/javascript/javascript-return-statement/>
13. <https://stackoverflow.com/questions/17337064/does-every-javascript-function-have-to-return-a-value>
14. <https://simpledev.io/lesson/arrow-functions-js/>